Deliverable 2
Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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This deliverable reports how the plan approved in Deliverable 1 was carried out, and it is written as a record of what was tested and how. Its thesis is that patch-based tokenisation makes certain frequencies poorly observable; that the affected set is fixed by the patch and the stride and splits into two branches the design keeps apart; and that the effect is measurable, partly mitigable by design, and representable in the semantics a control agent acts on. What follows runs from the trained models, to the frequency behaviour measured on them, to the Bayesian analysis that tests it, and then to the semantic layer and the security reading of the result.

Phase-Locking and Structural Aliasing

The two terms name different things, and separating them is what makes the hypotheses falsifiable rather than definitional. Phase-locking is a property of the input signal and the tokenisation geometry alone. It is exact arithmetic on (f, f_s, P, S) and holds for any model, trained or not. Deliverable 1 derives it as two conditions: $x[n+S] = x[n]$, under which consecutive patches are identical, and $x[n+P] = x[n]$, under which every patch holds a whole number of cycles. We call a frequency satisfying the first a stride lock and one satisfying the second a patch null, and the two sets they generate the stride branch and the patch branch of $\mathcal{F}_{lock}$. The set is their union, so a frequency may belong to one branch or to both. Structural aliasing is an empirical property of the learned representations, in which distinct inputs are mapped to nearly identical latent states and discriminative information is lost. Phase-locking makes it possible; it does not make it happen.

The split into two branches has a sharp experimental consequence. $H3$ makes a separate claim about each branch, so evidence has to be attributable to one of them, and that requires frequencies where only one condition holds. When S divides P the stride branch is contained in the patch branch: every locked frequency then satisfies both, and no observation in that geometry can say which one produced a dip. Only $S \nmid P$ yields stride-only sites, and supplying them is what the design of Table 1 is built for.

Models

Table 1 lists the nineteen geometries the analysis runs on. All are trained from scratch under the same 100k-step budget and from the same seed, so that no comparison is confounded by training length, and all

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

P	S	O	**tokens**	$S \mid P$
8	8	0.000	256	yes
16	8	0.500	255	yes
16	12	0.250	170	**no**
16	15	0.062	136	**no**
16	16	0.000	128	yes
24	8	0.667	254	yes
24	12	0.500	169	yes
24	15	0.375	135	**no**
24	16	0.333	127	**no**
24	20	0.167	102	**no**
24	24	0.000	85	yes
32	8	0.750	253	yes
32	12	0.625	169	**no**
32	15	0.531	135	**no**
32	16	0.500	127	yes
32	20	0.375	101	**no**
32	24	0.250	85	**no**
32	$28^{\dagger}$	0.125	73	**no**
32	32	0.000	64	yes

Table 1: *The nineteen geometries, all trained from scratch (seed* 42*) under the same* 100*k-step budget.* $O = (P - S)/P$*; the token count is at the* 2048*-sample training context and is what the overlap costs.* $S \mid P$ *decides whether the geometry can separate the two lock branches at all; Table 2 counts what each one supplies.* † *This stride does not divide the probing context: the run is swept but not fitted.*

are published as one repository with a subfolder per geometry [1]. The repository also carries runs at the shortest strides; those are excluded here, matching Deliverable 1, because their stride-lock class has too few in-band members to carry a local $H1$ contrast. The published `chronos-bolt-tiny` checkpoint is not used at any point [2], so the P=S=16 baseline is budget-matched like the rest of the design.

The design is not symmetric for its own sake; each axis identifies something. Ten runs have a stride grid that is not contained in their patch grid, and between them they supply the 68 stride-only sites, the only frequencies at which a behavioural loss is attributable to the stride branch alone. Six strides appear at more than one patch size, which is what identifies the patch branch of $H3$ and makes δ_P estimable. The overlap ratio spans 0 to 0.75, and three runs share $O = 0.5$ at three different patch sizes, which is what separates an effect of the overlap ratio from an effect of the absolute stride. The token count in the fourth column is the price of that overlap: at a fixed context it grows as $1/S$, so the widest-overlap run carries four times the sequence length of the P=S=32 baseline.

One run is treated differently. The probing context has to close on a whole number of strides, or the last patch is internally padded and the padding, not the geometry, produces the collapse. That holds for every stride but the one marked with a dagger, whose smallest admissible context is far outside the probing convention: it is swept and reported descriptively, not fitted.

Table 2 states the same design arithmetically. Each branch is the harmonics of a single spacing, so the spacing is the branch, and the only question left is where the two combs meet: on a third comb, spaced f_s/g with $g = \gcd(P, S)$. What survives once those shared members are removed is the stride-only column, zero exactly when $S \mid P$, and its total over the design is the 68 sites above.

$P = 8$						$f_s/P = 64\,\text{Hz}$
Model	f_s/S [Hz]	in band	f_s/g [Hz]	shared	**stride-only**	$\lvert\mathcal{F}_{lock}\rvert$
`p8-s8`	64	3	64	3	0	3

$P = 16$						$f_s/P = 32\,\text{Hz}$
Model	f_s/S [Hz]	in band	f_s/g [Hz]	shared	**stride-only**	$\lvert\mathcal{F}_{lock}\rvert$
`p16-s8`	64	3	64	3	0	7
`p16-s12`	42.67	5	128	1	**4**	11
`p16-s15`	34.13	7	512	0	**7**	14
`p16-s16`	32	7	32	7	0	7

$P = 24$						$f_s/P = 21.33\,\text{Hz}$
Model	f_s/S [Hz]	in band	f_s/g [Hz]	shared	**stride-only**	$\lvert\mathcal{F}_{lock}\rvert$
`p24-s8`	64	3	64	3	0	11
`p24-s12`	42.67	5	42.67	5	0	11
`p24-s15`	34.13	7	170.67	1	**6**	17
`p24-s16`	32	7	64	3	**4**	15
`p24-s20`	25.6	9	128	1	**8**	19
`p24-s24`	21.33	11	21.33	11	0	11

$P = 32$						$f_s/P = 16\,\text{Hz}$
Model	f_s/S [Hz]	in band	f_s/g [Hz]	shared	**stride-only**	$\lvert\mathcal{F}_{lock}\rvert$
`p32-s8`	64	3	64	3	0	15
`p32-s12`	42.67	5	128	1	**4**	19
`p32-s15`	34.13	7	512	0	**7**	22
`p32-s16`	32	7	32	7	0	15
`p32-s20`	25.6	9	128	1	**8**	23
`p32-s24`	21.33	11	64	3	**8**	23
`p32-s28`†	18.29	13	128	1	**12**	27
`p32-s32`	16	15	16	15	0	15

Table 2: *Where each geometry's locked frequencies sit, in band $[2, 250]$ Hz at $f_s = 512$ Hz. Each branch is the harmonics of one spacing, so the spacing is the whole branch: the stride branch is the multiples of f_s/S, the patch branch the multiples of f_s/P, given once per group;* in band *counts the stride branch's members. The two combs meet on a third, spaced f_s/g with $g = \gcd(P, S)$, which is what the shared column counts; a spacing above the band means they never meet inside it and nothing is shared.* ***Stride-only*** *is what is left of the stride branch once the shared members are removed, and it is zero exactly when $S \mid P$: those are the only frequencies at which a loss is attributable to the stride alone. $\lvert\mathcal{F}_{lock}\rvert$ is the union of the two branches, not the sum. † This stride does not divide the probing context: the run is swept but not fitted.*

Observed Frequency Behaviour

Every geometry is swept one hertz at a time over $[2, 250]$ Hz at $f_s = 512$ Hz, on a context of 480 samples with a 64-step horizon. The sweep grid is the uniform 1 Hz grid unioned with the exact predicted sites of every geometry, so no model is scored only where its own arithmetic predicts a dip and the non-integer sites are covered. Each swept frequency is measured three ways: the frequency the model regenerates, the amplitude it recovers relative to the injected one, and whether the input frequency can still be read back from the internal state by a probe. Locks are compared against two controls sharing the same background realisation and the same phase, placed at the largest offset not exceeding $0.25 f_s/S$ that keeps both clear of either branch. As in Deliverable 1 the tone rides on 100 TSMixup and 100 KernelSynth signals normalised to unit variance, and the pure-sinusoid sweep is kept as the exact-degeneracy reference.

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[TO FILL, Collaborating author: the figure pages, from the current run of `reconstruction_figures.ipynb`. One full-width page per patch size, grouping that size's strides so that a feature moving between rows belongs to the stride branch; then the two cross-geometry summaries the notebook writes last. Each caption should state what the two panels plot, what changes between rows, and that the dashed lines are $\mathcal{F}_{lock}$ and the colour scale is probe accuracy from chance to one.]

Bayesian Analysis Design

Deliverable 1 applied a two-part framework to $H1$ alone. Here every claim is given its own measurement, its own model, and its own estimand: the single quantity whose posterior decides it. Table 3 lists the models with the value each claim predicts, Table 4 every prior, and Table 5 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Model	**Claim**	**Estimand**	**Predicts**	**Input**
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 3: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to. $H1$ is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible. $H3$ is split into its two branches, since $\mathcal{F}_{lock}$ is a union and a frequency may belong to either. $M1$ is not a separate model but the configuration level of A, promoted to a named estimand.*

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 4 gives every prior with the role its parameter plays and the range over which the prior scale is varied as a sensitivity check; all effect priors are centred on zero effect, so no claim can be inherited from them. The fit returns a distribution rather than a point estimate, which is what permits the third verdict, inconclusive, that a pass/fail test cannot give.

$H1$

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast is unchanged from Deliverable 1,

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{1}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{2}$$

Three offsets build μ_i, one per configuration, lock harmonic and background draw, so that an unusual frequency or a single corpus cannot pass itself off as the effect. The heavy tails are the one deliberate departure from a Gaussian. Recovery falls to near zero where the model rebuilds nothing, and the logarithm then returns very large contrasts; under a Gaussian a few of them would set the answer alone.

The last term is the only unusual one, and it defines the estimand. Writing β_c as a draw from a distribution centred on $\bar{\beta}$ states that the geometries are instances of one phenomenon rather than unrelated experiments, and two things follow. First, $\bar{\beta}$ exists as a parameter: $H1$ is a claim about patch-based tokenisation, not about one checkpoint, so it needs a coefficient standing for the typical effect and carrying its own uncertainty. Second, each geometry weighs one unit. The number of usable triplets varies several times over with how many lock sites fall in band, so pooling every row into one coefficient would let the largest design, not the largest effect, decide the answer. Nineteen configurations make that level worth fitting, and let δ_O and δ_P be estimated together instead of confounded. The overlap enters as a ratio and the patch size as $\log P$, because the patch grid has spacing f_s/P, so equal steps in $\log P$ are equal ratios of spacing. A tilde marks a covariate centred on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which is what leaves $\bar{\beta}$ the effect at the average configuration instead of an extrapolation to $O = 0$ and $P = 1$.

Model B asks the same question of the representation, on a frequency-local prequential codelength: at each candidate f_c a probe must separate tones at $f_c \pm 1\,\text{Hz}$ from a captured internal state. Ten states are

Model	**Parameter**	**Prior**	**Role**
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 4: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 1 makes estimable. All are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect. Prior scales are varied over $\{0.25, 0.5, 1.0\}$ as a sensitivity check.*

probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head. Chronos-Bolt carries a [REG] token in the encoder only, its decoder being driven by a single decoder-start token, so the decoder-side vectors are named decoder states. The band tasks of Deliverable 1 each summarise hundreds of frequencies at once, leaving the lock indicator nothing to attach to, and are kept as a cross-check. The regression is Gamma with a log link; $e^{\theta_{lock}}$ reads as how many times more bits a locked frequency costs, and probe stage and geometry enter as zero-mean offsets, since codelengths differ several-fold between stages.

*H*2

$H2$ uses the same triplets with one change. Equation 1 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{3}$$

The estimand σ_ϕ is the spread of those eight offsets, and it answers the question with one number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone decides it, they are equal and σ_ϕ is near zero, as $H2$ predicts. Slices are used rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out separates the size of the phase effect from the evidence that it exists.

Claim	**Rule**	**Prior**	**Reads as**
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 5: *Decision rules, fixed before any posterior was inspected.* ***Prior*** *is the probability each rule already scored under the priors of Table 4: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive. Support and refutation are separate rules, since failing to establish a* 20% *attenuation does not establish that there is none.*

$H3$

$H3$ was stated in Deliverable 1 as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}$ Hz for $S{=}12$ are covered.

The first question is where the dips sit. Each swept frequency carries two labels, one per branch:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \mathrm{OnStrideGrid} + \theta_P \cdot \mathrm{OnPatchGrid},\ \sigma)\,. \tag{4}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. $H3$ predicts both θ_S and θ_P negative, so the two-label fit should win the leave-one-out comparison against each label alone and against neither. The labels are not rivals for the same dips, since a frequency may carry either or both. But the two coefficients separate only where exactly one label applies, which for θ_S means the ten $S \nmid P$ geometries.

The second question is whether the dips move. A good fit above is still compatible with dips that happen to sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental of branch F is then compared with the spacing that branch predicts, $\Delta_g^S = f_s/S_g$ or $\Delta_g^P = f_s/P_g$:

$$\hat{f}_{1,g}^{F} \sim \mathcal{N}\left(\kappa_F\, \Delta_g^F,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{5}$$

The floor Δf is the sweep step, since a spacing read off a discrete grid is no more precise than one step. Without it, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

H3a The stride branch. *H3a* predicts $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that stays put while S changes refutes it. It is identified by the series at fixed P, which the design supplies at every patch size in the table.

H3b The patch branch. *H3b* predicts $\kappa_P = 1$ in the same way. It is identified by the pairs that share a stride and differ in patch size, which the design supplies at six distinct strides.

Decisions and validation

Every threshold in Table 5 was fixed before a posterior was inspected, and validation comes before interpretation. Prior predictive checks precede collection; a parameter-recovery check then simulates data with a known effect and confirms the model finds it, which is what separates "no effect" from "a design that cannot see one"; posterior predictive checks are stratified by frequency, phase, P, S and generator. Competing formulations are compared by leave-one-out cross-validation rather than by null-hypothesis testing. Sampling uses NUTS with four chains, and a fit is reported only at $\hat{R} < 1.01$, effective sample sizes above 1000, and no divergences.

Mitigation Strategy

The mitigation carried through the design is increased patch overlap, that is a shorter S at fixed P. Its mechanism is arithmetic: the stride branch has its lowest in-band member at f_s/S, so halving the stride doubles that frequency and thins the family. The same arithmetic bounds what it can achieve. Overlap does not touch the patch branch at all, so it thins at best one of the two and leaves the P/k-minute blind periods of the photovoltaic mapping in place, and at a fixed context the token count grows as $1/S$, which is a real constraint for an edge-deployed forecaster. Both halves are testable rather than asserted, and $M1$ in Table 5 is how. Alternative mitigations, including stride randomisation, coprime-stride ensembling and phase-aware patch embeddings, are evaluated in the final deliverable.

Results

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Semantic Layer

The ontology reproduced in Appendix A closes the loop from the signal to the decision. It represents the photovoltaic micro-grid, its telemetry and a single control agent, and it makes the tokenisation geometry a first-class object rather than a property of the model: a `pv:TimeSeriesSignal` has a `pv:PatchedRepresentation` carrying `pv:patchSize`, `pv:stride` and a derived `pv:overlapRatio`, and one signal may hold several, one per (P, S) under evaluation. Since the lock conditions are arithmetic on those same fields, the set of candidates is derivable from the knowledge base rather than hard-coded beside it, which is what lets an agent recompute it when a configuration or a sampling interval changes. Phase-locking is what the ontology infers; structural aliasing is what it records.

Where the Bayesian analysis enters is the point that matters here. An `pv:ObservabilityAssessment` does not read a state off the geometry. It records the estimand, its effect size, its credible interval and its posterior probability, and assigns StructurallyAliased only to a frequency that is a lock candidate and also carries confirmed evidence of loss, $A = C \wedge B$ in the notation of Appendix A. Geometry alone yields a candidate, never a verdict. Because the confirmation threshold exceeds one half, confirmed loss and confirmed preservation cannot both hold, and everything else falls to PartiallyObservable: an unfinished test, a credible interval crossing the threshold, or conflicting metrics. That is the inconclusive verdict of Table 5 carried into the semantics, and its consequence is operational rather than documentary, because the assessment selects the control mode. Confirmed aliasing forces the safe mode and suspends forecast-driven battery decisions; a merely partially observable assessment reduces the battery cap and lowers decision confidence. Uncertainty about the representation is therefore propagated to the actuator instead of being resolved by assumption.

Security Considerations

The forecaster is not a standalone artefact. It sits inside a control loop in which its output drives battery dispatch and curtailment on a photovoltaic micro-grid, so a forecasting failure is a control failure, and the property under test in this deliverable has to be read as a security property and not only as an accuracy one. The NIST AI Risk Management Framework [3] is the reference we adopt for that reading, because it separates the characteristics we can evidence here, validity and reliability, security and resilience, from the governance functions that turn evidence into a control, and because it asks for known limitations to be mapped and measured rather than merely disclosed.

The limitation itself is unusual, and naming its shape correctly is what makes the rest follow. Structural aliasing is deterministic, attacker-independent and present at deployment time. It is not introduced by an adversary and it is not a defect in the training data; it is a consequence of the tokenisation geometry, and it is closed-form. The set of affected frequencies follows from (f_s, P, S) alone, and those three numbers are configuration, not weights. An adversary therefore needs no model access, no gradients, no query budget and no knowledge of the training corpus to know exactly where the model is blind. In the taxonomy of adversarial machine learning [4] this places the weakness with the availability violations rather than with integrity or privacy, and it places the attack in the white-box-free category, which is the part a conventional threat model tends to miss because it reasons about perturbations of the input tensor rather than of the physical process.

The attack surface follows directly. The perturbation an adversary needs is physical and small: a periodic component injected at, or dithered onto, a locked frequency by any controllable load, inverter switching pattern or curtailment schedule is present in the plant and absent from the forecast, because the tokenisation maps it to the same latent state as its absence. Two consequences matter operationally. The signal is invisible rather than misread, so residual-based anomaly detection downstream of the forecaster does not see it either; and the same arithmetic that makes the exploit cheap makes it stealthy, since the injected component is legitimate plant behaviour at a frequency the operator has no reason to treat as special. Detection has to be placed before tokenisation, on the raw telemetry, or it is placed inside the blind spot it is meant to guard.

What the framework then requires is that this be mapped, measured and managed rather than noted. Mapping is Table 1 together with the lock arithmetic, which enumerates the affected set for every deployed configuration. Measurement is the Bayesian analysis of Bayesian Analysis Design: the estimand and its credible interval are the evidence artefact, and the prior probability recorded beside each threshold is what makes the verdict auditable rather than asserted. Management has two halves, one in the model and one in the system. In the model it is the overlap mitigation, whose bound is stated with it, since overlap thins one branch and leaves the other untouched, so residual risk survives by construction and has to be reported as such. In the system it is the semantic layer, where a PartiallyObservable assessment degrades actuation instead of failing silently, which is the resilience property the framework asks for. The full risk assessment, with the threat model, the residual-risk register and the operational controls, is carried in the final deliverable.

Appendix A: Ontology of the Photovoltaic Application Domain

This document defines a compact ontology for a photovoltaic system governed by one control agent. It connects environmental and electrical observations to anomaly recognition, Chronos-style patched representations, structural-aliasing evidence, and bounded power-routing decisions [5]. Routing is included only as the operational endpoint of the semantic model; the principal concern remains whether the representation preserves the information on which the agent relies.

Domain Description

The modeled system contains four energy assets: a photovoltaic array, battery storage, an aggregate internal load, and an external grid. The internal load is a passive sink representing a factory or a fleet of machines. Its demand is observed, but it has no agent, machine-level taxonomy, curtailment command, or scheduling model. The external grid is an available bidirectional boundary bus. It supplies residual demand and accepts residual production.

Exactly one `pv:ControlAgent` coordinates the four passive assets. It observes one synchronized control cycle, assigns environmental, anomaly, and observability states, selects a control mode, and emits one or more directional decisions. There is no peer-to-peer negotiation among assets.

The seven admissible transfer directions are

$$\begin{array}{ccc} \text{PV} \to \text{Load}, & \text{PV} \to \text{Battery}, & \text{PV} \to \text{Grid}, \\ \text{Battery} \to \text{Load}, & \text{Battery} \to \text{Grid}, & \text{Grid} \to \text{Load}, \\ & \text{Grid} \to \text{Battery}. & \end{array} \tag{6}$$

They satisfy the cycle-level balance

$$P_{\text{pv}} + P_{\text{bat,dis}} + P_{\text{grid,in}} = P_{\text{load}} + P_{\text{bat,ch}} + P_{\text{grid,out}}. \tag{7}$$

All terms denote non-negative magnitudes; import and export are therefore kept separate rather than encoded by an ambiguous sign.

Purpose and Scope

The ontology provides the smallest semantic layer needed to represent the physical boundary, validated observations, environmental context, three selected anomalies, and the effect of patching on model observability. It does not model individual machines, market prices, dispatch optimisation, grid faults, or a deployed real-time controller. Load demand, battery SoC, asset limits, and the optional grid-exchange reference are asserted or simulated because they are not present in the photovoltaic telemetry dataset.

The environmental states are operational generation proxies, not meteorological diagnoses. Likewise, the routing rules describe a deterministic conceptual policy rather than the central experimental contribution. Structural aliasing remains an empirically tested property of a signal-representation pair; no fixed synthetic frequency is transferred to the one-minute photovoltaic signal.

Ontology Representation

Core classes and closed vocabularies Table 6 defines the compact class interface. The four asset subclasses form a disjoint union; each `pv:PVSystem` contains exactly one asset of each type and is managed by exactly one control agent. Closed operational vocabularies are represented as named individuals rather than deeper class hierarchies.

Class	**Role**
`pv:PVSystem`	Aggregate system and scope of each control cycle.
`pv:EnergyAsset`	Parent of the four passive physical assets.
`pv:PhotovoltaicArray`	Intermittent renewable source measured through PV telemetry.
`pv:BatteryStorage`	Bounded storage asset that can charge, discharge, or hold.
`pv:InternalLoad`	Aggregate passive demand of a factory or machine fleet.
`pv:ExternalGrid`	Reliable bidirectional boundary bus for residual balancing.
`pv:ControlAgent`	The single decision-making entity.
`pv:SystemObservation`	Time-stamped snapshot of PV, environment, battery, load, and grid reference.
`pv:TimeSeriesSignal`	Physical signal and its temporal descriptors.
`pv:PatchedRepresentation`	A model input representation defined by patch and stride.
`pv:ObservabilityAssessment`	Evidence linking a signal-representation pair to an observability state.
`pv:GenerationContext`	Operational proxy for expected solar generation.
`pv:AnomalyState`	Detected physical or sensing abnormality.
`pv:ObservabilityState`	Semantic assessment of information retained after patching.
`pv:ControlMode`	Normal, conservative, or safe operating policy.
`pv:PowerRoute`	Closed vocabulary of the seven admissible source–destination pairs.
`pv:PowerRoutingDecision`	One directional transfer or hold decision with an optional bounded setpoint.

Table 6: *Core classes in the simplified ontology.*

The closed vocabularies are:

Generation context: ClearSky, PartialCloud, Overcast, and Nighttime.
Anomaly: InverterTrip, RampEvent, and SensorMalfunction.
Observability: Observable, PartiallyObservable, and StructurallyAliased.
Control mode: NormalMode, ConservativeMode, and SafeMode.
Power route: PVToInternalLoad, PVToBattery, and PVToExternalGrid;
BatteryToInternalLoad and BatteryToExternalGrid;
ExternalGridToInternalLoad and ExternalGridToBattery.

Formally, each vocabulary class is an OWL-style enumeration of the listed named individuals (`oneOf`). The five vocabulary classes are pairwise disjoint, and all listed values are declared globally different (`AllDifferent`). This prevents a value such as SafeMode from being silently identified with ClearSky or NormalMode under open-world semantics. Each valid time-stamped observation has exactly one generation context. Each completed observability assessment has exactly one observability state, and each routing decision records exactly one control mode. Co-issued decisions based on the same synchronized observation must record the same mode. Anomalies are not declared mutually exclusive because, for example, an abrupt inverter trip also creates a large power change. Deterministic control is obtained through the priority rules in the closed-loop section.

Object properties The semantic relations in Table 7 connect the physical boundary to the inference and control layers. Each active `pv:PowerRoutingDecision` has exactly one route value, one source, and one distinct destination. Its non-negative setpoint magnitude is optional because `Balance` may denote passive residual exchange. A cycle that simultaneously serves the load, charges the battery, and exports a residual uses three instances of the same decision class, all based on the same observation and mode.

Property	**Domain**	**Range or meaning**
`pv:hasAsset`	`pv:PVSystem`	`pv:EnergyAsset`
`pv:managedBy`	`pv:PVSystem`	exactly one `pv:ControlAgent`
`pv:hasObservation`	`pv:PVSystem`	`pv:SystemObservation`
`pv:hasSignal`	`pv:PhotovoltaicArray`	`pv:TimeSeriesSignal`
`pv:hasRepresentation`	`pv:TimeSeriesSignal`	`pv:PatchedRepresentation`
`pv:assessesRepresentation`	`pv:ObservabilityAssessment`	exactly one `pv:PatchedRepresentation`
`pv:hasGenerationContext`	`pv:SystemObservation`	zero or one `pv:GenerationContext` value
`pv:hasAnomaly`	`pv:SystemObservation`	zero or more `pv:AnomalyState` values
`pv:hasObservabilityState`	`pv:ObservabilityAssessment`	one `pv:ObservabilityState` value
`pv:selectsMode`	`pv:PowerRoutingDecision`	exactly one `pv:ControlMode` value
`pv:makesDecision`	`pv:ControlAgent`	`pv:PowerRoutingDecision`
`pv:basedOn`	`pv:PowerRoutingDecision`	observation, assessment, or inferred state
`pv:hasRoute`	`pv:PowerRoutingDecision`	zero or one `pv:PowerRoute` value
`pv:hasSourceAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`
`pv:hasDestinationAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`

Table 7: *Object-property signatures.*

`pv:hasRepresentation` is intentionally non-functional because one signal can be evaluated under several (P, S) configurations. Each assessment identifies exactly which representation it evaluates. Each decision is based on exactly one current observation and its completed assessment; its own `pv:selectsMode` value therefore scopes the mode without accumulating incompatible modes on the persistent agent. The range of `pv:basedOn` is the union of observation and inferred-assessment entities, rather than the intersection that multiple RDFS range declarations would imply.

OWL's open-world semantics do not by themselves validate that endpoint types match a named route. The operational validator therefore admits only the seven ordered pairs in Equation 6, rejects identical endpoints, and checks consistency among `pv:hasRoute`, source, and destination. A hold decision has no route or endpoints and has a zero or absent setpoint. The `pv:basedOn` relation is retained only as the minimum evidence link required to justify a selected mode; no separate trace model is introduced.

Data properties and units The synchronized observation combines the measured photovoltaic and environmental variables with asserted or simulated battery, load, and grid-reference values. The control-critical quantities are PV power P_{pv}, tilted irradiance G_{tilt}, normalized battery state of charge $s \in [0, 1]$, and aggregate load demand $P_{\mathrm{load}} \geq 0$. Each has a separate validity flag derived from presence, finiteness, freshness, and a configured physical range. Horizontal irradiance, air temperature, wind speed, and wind direction remain environmental evidence but do not independently block the controller. The signed `pv:gridExchangeReferenceW` is optional, with positive values denoting export; it is usable only when present, finite, and fresh.

Missing or stale data cannot be inferred safely from OWL's open-world assumption alone. An operational validation layer therefore compares the observation time stamp with a configurable staleness limit, validates numeric ranges, and asserts the four critical Boolean flags consumed by the ontology. This keeps data-quality validation distinct from semantic classification.

Owner	Data properties
`pv:SystemObservation`	`pv:pvPowerW`, `pv:powerRampRateWPerS`, `pv:loadDemandW`, optional `pv:gridExchangeReferenceW`, `pv:tiltedIrradianceWm2`, `pv:horizontalIrradianceWm2`, `pv:airTemperatureC`, `pv:windSpeedMs`, `pv:windDirectionDeg`, `pv:batterySoC`, `pv:observationTimestamp`, `pv:pvPowerValid`, `pv:tiltedIrradianceValid`, `pv:batterySoCValid`, and `pv:loadDemandValid`.
`pv:BatteryStorage`	`pv:capacityWh`, `pv:reserveSoC`, `pv:targetSoC`, `pv:maximumSoC`, `pv:maximumChargePowerW`, and `pv:maximumDischargePowerW`.
`pv:TimeSeriesSignal`	`pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:periodSamples`, `pv:amplitudeW`, `pv:timeSpanSeconds`, and `pv:phaseOffsetSamples`.
`pv:PatchedRepresentation`	`pv:patchSize`, `pv:stride`, and derived `pv:overlapRatio`.
`pv:ObservabilityAssessment`	`pv:analysisStatus`, `pv:assessmentMethod`, `pv:assessmentMetric`, `pv:strideLockingRatio`, `pv:patchLockingRatio`, `pv:lockingTolerance`, `pv:effectEstimate`, `pv:posteriorLossProbability`, `pv:credibleIntervalLower`, `pv:credibleIntervalUpper`, and `pv:assessmentTimestamp`.
`pv:PowerRoutingDecision`	optional non-negative `pv:powerSetpointW`, `pv:decisionConfidence`, `pv:batteryCommand` in {Charge, Discharge, Hold}, `pv:pvCommand` in {Connect, Isolate}, and `pv:gridCommand` in {Import, Export, Balance}.

Table 8: *Quantitative property groups and units encoded by their names.*

A valid patched representation satisfies $P \geq 1$ and $1 \leq S \leq P$. Overlap is descriptive metadata:

$$\gamma = \frac{P - S}{P}. \tag{8}$$

It is not used as a standalone rule for declaring aliasing.

Inference Rules

Environmental generation context Let $g_0 < g_1 < g_2$ be site-configurable tilted-irradiance thresholds and let G_t denote the validated value of G_{tilt} at time t. The active generation context is

$$C_G(G_t) = \begin{cases} \mathsf{Nighttime}, & G_t \leq g_0, \\ \mathsf{Overcast}, & g_0 < G_t < g_1, \\ \mathsf{PartialCloud}, & g_1 \leq G_t < g_2, \\ \mathsf{ClearSky}, & G_t \geq g_2. \end{cases} \tag{9}$$

Values $g_0 = 0$, $g_1 = 150$, and $g_2 = 800\ \mathrm{W\,m^{-2}}$ are retained only as an illustrative configuration compatible with the current coursework. They are not universal meteorological boundaries. In particular, low irradiance can also be caused by solar position; the four values must therefore be read as generation-context proxies rather than certain weather diagnoses.

Anomaly detection and precedence Let $v_P(t)$, $v_G(t)$, $v_s(t)$, and $v_L(t)$ denote the validity of PV power, tilted irradiance, battery SoC, and aggregate load demand. The control-critical validity predicate is

$$V_t = v_P(t) \wedge v_G(t) \wedge v_s(t) \wedge v_L(t). \tag{10}$$

If V_t is false, no physical anomaly or forecast-driven route is inferred from that observation. A valid irradiance value may still support a descriptive generation context, but the operational validator asserts SensorMalfunction and the cycle proceeds directly to safe-mode selection. Within a validated cycle, absence of the three listed anomaly values is interpreted locally as "none detected"; this is a scoped closed-world policy, not a global OWL assumption. The signed ramp rate, measured in watts per second when Δt is expressed in seconds, is

$$r_t = \frac{P_t - P_{t-1}}{\Delta t}. \tag{11}$$

The anomaly rules are expressed with configurable thresholds: g_{trip} for sufficient irradiance, p_0 for near-zero power, and $r_{\max}$ for maximum admissible ramp magnitude.

$$\begin{aligned} \neg V_t &\longrightarrow \textsf{SensorMalfunction}, && (12)\\ V_t \wedge G_t \geq g_{\text{trip}} \wedge |P_t| \leq p_0 &\longrightarrow \textsf{InverterTrip}, && (13)\\ V_t \wedge V_{t-1} \wedge |r_t| > r_{\max} &\longrightarrow \textsf{RampEvent}. && (14) \end{aligned}$$

No numerical value is inherited for $r_{\max}$ because it must be calibrated after invalid values and outliers have been excluded. The controller evaluates sensor validity first. A sensor malfunction therefore cannot be reinterpreted as a physical ramp or inverter trip. If a valid inverter trip also satisfies the ramp predicate, the safe response to the trip takes precedence over ramp smoothing.

Signal observability and structural aliasing For a signal frequency f, sampling frequency f_s, patch size P, and stride S, define a stride-lock ratio and a patch-periodicity ratio

$$\rho_S = \frac{fS}{f_s}, \qquad \rho_P = \frac{fP}{f_s}. \tag{15}$$

A stride phase-lock candidate L_S and a within-patch repetition candidate L_P are

$$\begin{aligned} L_S &= \left[\exists m \in \mathbb{Z}^+ : |\rho_S - m| \leq \varepsilon_S\right], \\ L_P &= \left[\exists k \in \mathbb{Z}^+ : |\rho_P - k| \leq \varepsilon_P\right], \\ C &= (L_S \vee L_P) \wedge \left(0 < f < \frac{f_s}{2}\right), \end{aligned} \tag{16}$$

where ε_S and ε_P are tied to frequency resolution and are never hardcoded frequency bands. Only L_S expresses equality of consecutive patch phases; L_P records whole-cycle repetition inside a patch and is retained as an H3 experimental factor. Their union C is a candidate geometry class, not proof that two domain concepts are indistinguishable.

This is a division of labour, and it bounds what the ontology can conclude on its own. Phase-locking is *derivable*: L_S, L_P and C follow by rule from `pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:patchSize` and `pv:stride`, all of which the knowledge base already holds, so a reasoner can enumerate the candidate set for any configuration without observing a model. Structural aliasing is *not* derivable and is only recorded: it enters through B below, which is posterior evidence from an experiment external to the ontology, and no rule here can produce it. An agent using this ontology therefore detects phase-locking and reports structural aliasing.

Let q denote the task-relevant information being tested, for example frequency or amplitude. Let $\Delta_{\text{loss}}(q)$ be its predeclared behavioral or representational loss relative to matched controls, δ the maximum practically negligible loss, D the experimental evidence, and $\tau > 1/2$ the posterior confidence threshold. Define confirmed information loss B and confirmed preservation R as

$$B = \left[\Pr(\Delta_{\text{loss}}(q) > \delta \mid D) \geq \tau\right], \qquad R = \left[\Pr(\Delta_{\text{loss}}(q) \leq \delta \mid D) \geq \tau\right]. \tag{17}$$

Because $\tau > 1/2$, B and R cannot both hold for the same assessment. The assessment interface is designed to receive the planned offline evidence already specified in Deliverable 1: a local amplitude-recovery contrast with a heavy-tailed Student-t_4 likelihood and a Gamma regression with log link for non-negative prequential-MDL codelength. The population contrast uses the weakly informative prior $\bar{\beta} \sim \text{Student-}t_4(0, 0.5)$; the Gamma model uses $\theta_{\text{lock}} \sim \mathcal{N}(0, 0.5)$ and shape $k \sim \text{Gamma}(2, 0.1)$. The stored record identifies the tested (P, S) configuration, matched controls, method and metric, effect estimate, 95% credible interval, and posterior probability of at least 20% attenuation. These are evidence fields, not universal ontology thresholds. Until that planned inference has completed and satisfies the predeclared criterion, no instance may be asserted as StructurallyAliased.

The three observability values are assigned as follows:

$$C_O = \begin{cases} \textsf{StructurallyAliased}, & A, \\ \textsf{Observable}, & \neg A \wedge R, \\ \textsf{PartiallyObservable}, & \neg A \wedge \neg R, \end{cases} \qquad A = C \wedge B. \tag{18}$$

Observable therefore means that positive evidence shows the tested representation preserves the task-relevant information within tolerance δ. It does not mean global observability of the physical system, nor does it merely mean that aliasing was not found. Candidate geometry may still be Observable when the experiment demonstrates preservation. PartiallyObservable is the epistemic fallback: preservation has not been certified, while structural aliasing has not been confirmed. It includes a candidate with inconclusive evidence, an unfinished test, a credible interval crossing the decision threshold, conflicting metrics, or no preservation evidence. The term means *partially trusted*, not that a fixed fraction of samples or sensors is visible.

H1 compares each candidate frequency with matched nearby non-candidate controls and predicts higher MDL codelength or lower amplitude recovery; absent or non-localized loss refutes H1. H2 tests whether the loss at a candidate frequency remains approximately stable across sampled phase offsets; systematic phase dependence refutes that hypothesis. H3 tests whether candidate degradation sites move proportionally to $1/S$ or $1/P$ when the corresponding parameter varies. These hypotheses contribute controlled evidence to D, but remain falsifiable experimental factors rather than ontology subclasses or automatic action rules. Sensor validity is modeled separately as an anomaly and never relabels a model representation as structurally aliased.

When an experimental generator provides a phase ϕ in radians, the corresponding temporal offset is converted before storage:

$$o_{\text{samples}} = \frac{\phi f_s}{2\pi f}. \tag{19}$$

Neither ϕ nor o_{samples} is compared directly with stride to infer an anomaly.

The semantic propagation path is explicit:

$$\begin{aligned} \texttt{pv:TimeSeriesSignal} &\longrightarrow \texttt{pv:PatchedRepresentation} \\ &\longrightarrow \texttt{pv:ObservabilityAssessment} \\ &\longrightarrow \texttt{pv:ControlMode} \longrightarrow \texttt{pv:PowerRoutingDecision}. \end{aligned} \tag{20}$$

Aliasing can therefore reduce confidence in forecast-derived production, ramp expectations, forecast-driven anomaly recognition, and routing. It does not alter the directly observed values of G_{tilt}, P_{pv}, load demand, or SoC; those values are governed by the separate validity path.

Closed-Loop Control

Mode selection The agent evaluates each synchronized control cycle using the strict priority

$$\textsf{SafeMode} \succ \textsf{ConservativeMode} \succ \text{RampOverride} \succ \text{NormalRouting}. \tag{21}$$

Let A_{safe} mean that SensorMalfunction or InverterTrip is present. The selected mode is

$$M_t = \begin{cases} \textsf{SafeMode}, & A_{\text{safe}} \vee (C_O = \textsf{StructurallyAliased}), \\ \textsf{ConservativeMode}, & C_O = \textsf{PartiallyObservable}, \\ \textsf{NormalMode}, & \text{otherwise}. \end{cases} \tag{22}$$

In SafeMode, the battery command is Hold and the grid balances the internal load. The PV command is Isolate only for an inverter trip; for sensor malfunction or confirmed structural aliasing it remains connected, but forecast- and ramp-driven battery decisions are suspended. When load demand itself is invalid, Balance carries no invented numerical setpoint. In ConservativeMode, the same seven route types remain available with a configurable reduced battery-power cap, normal target/reserve bounds, and low decision confidence. A concurrent ramp cannot activate either extended ramp bound; this implements the precedence in Equation 21.

Routing policy The ordinary policy serves the internal load before charging the battery or exporting energy. Let

$$P_{\mathrm{sur}} = (P_{\mathrm{pv}} - P_{\mathrm{load}})^{+}, \qquad P_{\mathrm{def}} = (P_{\mathrm{load}} - P_{\mathrm{pv}})^{+}. \tag{23}$$

Here *daylight* abbreviates {ClearSky, PartialCloud, Overcast}. A cycle issues a route only for a strictly positive transfer and may issue several co-mode decisions. All setpoints are clipped to SoC, capacity, and charge/discharge limits.

Figure 1 shows the priority scan and Table 9 states its boundary cases. The conservative branch rejoins the ordinary load-first router under its reduced cap. The ramp branch is available only in NormalMode.

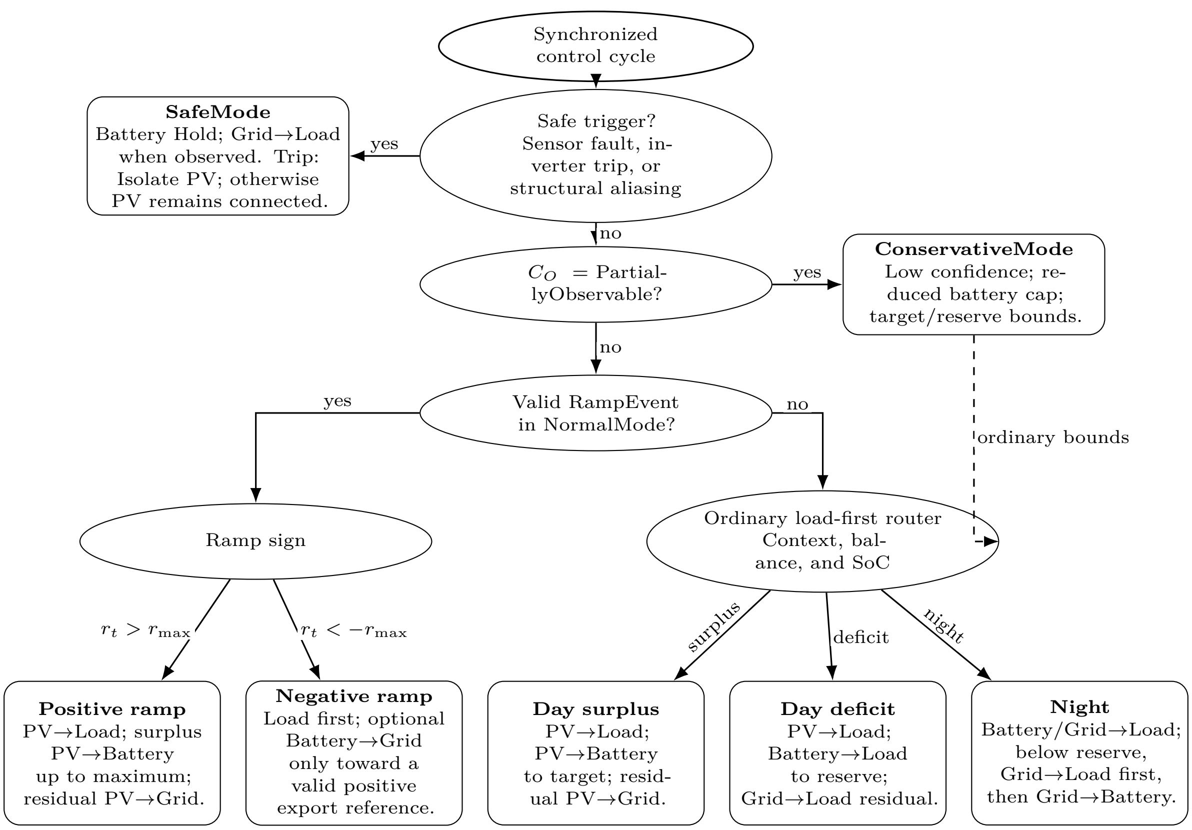

Figure 1: *Load-first routing decision tree. Elliptical bubbles are ordered gates; rounded bubbles are terminal policies. Safe and conservative decisions precede the normal-mode ramp override, while the ordinary router covers daylight surplus, daylight deficit, and nighttime operation.*

Condition	**Ordered route set**	**Constraint**
Daylight surplus	PV→Load; PV→Battery; PV→Grid	Serve demand, charge to target, then export the residual.
Daylight deficit	PV→Load; Battery→Load; Grid→Load	Discharge only to reserve; the grid covers the remaining deficit.
Night, $s > s_{\text{reserve}}$	Battery→Load; Grid→Load	Battery serves demand to reserve; the grid supplies the residual.
Night, $s \leq s_{\text{reserve}}$	Grid→Load; if $s < s_{\text{reserve}}$, Grid→Battery	Supply the load first; charge only below reserve and stop at reserve.
Positive ramp with surplus	PV→Load; PV→Battery; PV→Grid	NormalMode may charge beyond target, never beyond maximum SoC.
Negative ramp with valid positive reference	load-serving routes; Battery→Grid	Export only after serving the load, within residual discharge capacity and above reserve.
Negative ramp without usable reference	ordinary surplus/deficit routes	Missing, stale, non-finite, zero, or negative reference disables export catch-up.
ConservativeMode	ordinary load-first routes	Reduced battery cap; target/reserve bounds; no ramp extensions.
SafeMode	Grid→Load when observed	Battery Hold; grid Balance; isolate PV only for inverter trip.

Table 9: *Routing matrix for the seven admissible transfer directions.*

For a negative ramp, Battery→Grid is admissible only after the internal load has been fully served, the export reference is present, fresh, finite, and positive, measured export is below that reference, the battery remains above reserve, and discharge capacity remains after load support. If any condition fails, the ordinary load-first route applies. At night, Grid→Battery begins only after Grid→Load has been satisfied and stops when reserve is restored. These rules preserve Equation 7 without turning the routing policy into a separate optimisation problem.

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